

Math 421

Friday, October 10

Announcements

- Homework 5 due **today**
- Exam 1 – 10/15 – 5:45-7:15pm
 - If you have a conflict, email me (if you haven't already) and **fill out makeup exam request form** (linked from Exam Information page on Canvas) ASAP!
 - If you are a McBurney student, sign up to take the exam at the testing center through RegisterBlast.

9.7 Theorem (Basic Examples)

- a) $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.
- b) $\lim a^n = 0$ if $|a| < 1$.
- c) $\lim (n^{1/n}) = 1$.
- d) $\lim (a^{1/n}) = 1$ for $a > 0$.

Activity – Prove 9.7 a)

Theorem. $\lim \left(\frac{1}{n^p} \right) = 0$ for $p > 0$.

Discussion.

Proof.

Diverging to $\pm\infty$

Definition 9.8

- $\lim s_n = +\infty$ provided for each $M > 0$ there is a number N such that $n > N$ implies $s_n > M$.
- $\lim s_n = -\infty$ provided for each $M < 0$ there is a number N such that $n > N$ implies $s_n < M$.

Terminology: A sequence (s_n) has a **limit** or the **limit exists** if (s_n) converges to a real number or diverges to $\pm\infty$.

Example/Activity

Example: $\lim n^2 = +\infty$

Activity: $\lim(\sqrt{n} + 7) = +\infty$

Activity

1. What does it mean for an unbounded sequence to not diverge to $+\infty$ or $-\infty$, i.e. what is the negation of the definition?
2. Prove that the unbounded sequence $s_n = (-1)^n n$ does not have $+\infty$ as a limit. What proof technique should you use?
3. Modify your proof to show that $s_n = (-1)^n n$ does not have $-\infty$ as a limit.

Caution!

The limit laws we've previously stated do not apply to sequences that have infinite limits, we need to prove these separately.

Some Limit Laws for $\pm\infty$

Theorem 9.9. If $\lim s_n = +\infty$ and $\lim t_n > 0$, then $\lim s_n t_n = +\infty$.

Q: Why must $\lim t_n > 0$?

Theorem 9.10. $\lim s_n = +\infty$ if and only if $\lim \left(\frac{1}{s_n}\right) = 0$.

Divide & Conquer – Scratch Work

Theorem 9.10. $\lim s_n = +\infty$ if and only if $\lim \left(\frac{1}{s_n}\right) = 0$.

($\Rightarrow$) If $\lim s_n = +\infty$, then $\lim \left(\frac{1}{s_n}\right) = 0$.

($\Leftarrow$) If $\lim \left(\frac{1}{s_n}\right) = 0$, then $\lim s_n = +\infty$.

Chapter 2: Sequences

Section 10 (Part 1): Monotone Sequences

Monotonic Sequences

Definition. A sequence (s_n) of real numbers is called **monotonic** if it is increasing or decreasing.

- A sequence (s_n) of real numbers is called an **increasing sequence** if $s_n \leq s_{n+1}$ for all n .
- A sequence (s_n) of real numbers is called a **decreasing sequence** if $s_n \geq s_{n+1}$ for all n .

Activity – Give an example of the following

- An increasing sequence
- A decreasing sequence
- A non-monotonic sequence.

True/False

- ❖ Every convergent sequence is bounded.
- ❖ Every bounded sequence is convergent.
- ❖ Every convergent sequence is monotonic.
- ❖ Every monotonic sequence is convergent.

Bounded Monotonic $\Rightarrow$ Convergent

Theorem 10.2. *All bounded monotone sequences converge.*

Discussion.

Bounded Monotonic $\Rightarrow$ Convergent

Theorem 10.2. *All bounded monotone sequences converge.*

Proof.